

Substring Avoidance across Bases:

How long can a doubling sequence run before it is forced to display a pattern?

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Abstract

Start at 1 and repeatedly apply $x \mapsto bx + c$ with a digit $c \in \{0, \dots, b-1\}$; this generates every positive integer exactly once, along its base- b expansion. Fix a target base B and a digit-string n . We ask how long such a generation can proceed while the *base- B* representation of every value produced avoids n as a contiguous factor. For $(b, B) = (2, 10)$ this is precisely the longest binary string, with no leading zero, all of whose prefixes read in decimal avoid n .

We separate three quantities. The literal *shortest* string reaching a value that contains n is shown to be a triviality of base conversion (Proposition 1). The substantive objects are the longest avoider $L_{b,B}(n)$ and its true dual, the shortest *maximal* avoider $h_{b,B}(n)$. We give exact data for $(2, 10)$ with single-digit n , and our main structural result is a dichotomy governed by whether $\log b / \log B$ is rational. When b and B are multiplicatively dependent we exhibit explicit eventually-periodic words proving $L_{b,B} = \infty$ (Theorem 2); when they are independent we prove the opposite— $L_{b,B}(n) < \infty$ for every n (Theorem 5)—by showing that the leading base- B digits of any chain equidistribute and hence display every pattern. This completes a dichotomy governed entirely by whether $\log b / \log B$ is rational, and exposes the finiteness as an *arithmetic* phenomenon (incommensurable bases), in deliberate contrast to Thue–Morse-style word avoidance.

Finally we study a self-referential variant in which each term must avoid all of its own predecessors (Section 5). A short reduction (Proposition 5) ties its finiteness to the same avoidance question, single-digit starts are solved exactly, and we isolate the accumulating self-constraints—rather than avoidance of the start—as the mechanism that bounds the sequence.

1 The generation tree

Fix integers $b, B \geq 2$. We call b the *generation base* and B the *reading base*.

Definition 1. The *generation tree* T_b is the rooted infinite tree with root 1 in which each node x has the b children $bx, bx + 1, \dots, bx + (b - 1)$. A *generation string* of length k is a path $x_1, x_2, \dots, x_k$ from the root, equivalently a word $d_1 d_2 \dots d_k$ over $\{0, \dots, b - 1\}$ with $d_1 \neq 0$ and $x_i = \text{val}(d_1 \dots d_i) = \sum_{j \leq i} d_j b^{i-j}$.

Thus generation strings are exactly the base- b representations of positive integers, and a length- k path corresponds to a k -digit base- b number. For $b = 2$ the two operations are $x \mapsto 2x$ (append 0) and $x \mapsto 2x + 1$ (append 1): the “doubling” process of the title.

Let $\rho_B(x) \in \{0, \dots, B - 1\}^*$ denote the base- B digit word of x (most significant digit first, no leading zero for $x \geq 1$). A *pattern* is a nonempty word $n = n_1 \dots n_\ell$ over $\{0, \dots, B - 1\}$ with $n_1 \neq 0$; we identify it with the integer $\nu := (n)_B = \sum_i n_i B^{\ell-i}$. Patterns are exactly the base- B representations of positive integers, matching the convention $n \in \mathbb{Z}_{\geq 1}$.

Definition 2. An integer $x \geq 1$ is *good* (for n, B) if n is not a factor of $\rho_B(x)$, and *bad* otherwise. A generation string $x_1, \dots, x_k$ is *valid* if every x_i is good.

Because the parent of x in T_b is $\lfloor x/b \rfloor$, validity of a path is equivalent to all of its ancestors being good. Hence a node x lies on some valid path iff x and all of $\lfloor x/b \rfloor, \lfloor x/b^2 \rfloor, \dots, 1$ are good. The valid nodes form a subtree of T_b rooted at 1 (assuming 1 is good), which we call the *avoidance tree* $A = A_{b,B}(n)$.

Definition 3. We study three quantities.

- (i) $L_{b,B}(n) = \sup\{k : \text{a valid string of length } k \text{ exists}\} \in \mathbb{N} \cup \{\infty\}$, the *longest avoider*: the height of A .
- (ii) $S_{b,B}(n) = \min\{k : \text{a generation string of length } k \text{ ends at a bad value}\}$, the *shortest reach*.
- (iii) $h_{b,B}(n) = \min\{k : \text{a maximal valid string of length } k \text{ exists}\}$, the *shortest maximal avoider*, where a valid string is *maximal* if none of its b one-letter extensions is valid (equivalently the terminal node is a leaf of A).

For the motivating instance $(b, B) = (2, 10)$, $L_{b,B}(n)$ is the length of the longest binary string with no leading zero whose every prefix, written in decimal, avoids n . For example $L_{b,B}(1) = 0$ (the prefix $\rho_{10}(1) = 1$ already contains 1), while $L_{b,B}(2) = 6$ is witnessed by $111100 = \text{val}^{-1}(60)$ with prefix values 1, 3, 7, 15, 30, 60, none containing the digit 2.

2 The shortest problem is base conversion

We first dispatch the shortest-reach quantity, which turns out to carry no arithmetic content; the genuinely dual notion is $h_{b,B}$, treated in Section 4.

Lemma 1 (Reachability). *Every integer $m \geq 1$ is the value of exactly one generation string, namely $\rho_b(m)$, whose length is $\lfloor \log_b m \rfloor + 1$.*

Proof. Immediate: T_b realizes the standard base- b numeration, and the number of base- b digits of m is $\lfloor \log_b m \rfloor + 1$. $\square$

Proposition 1 (Shortest reach). *For every admissible pattern n ,*

$$S_{b,B}(n) = \lfloor \log_b \nu \rfloor + 1, \quad \nu = (n)_B.$$

Moreover $S_{b,B}(n)$ is exactly the depth of the shallowest bad node, and that node is reachable through good ancestors.

Proof. The smallest bad integer is ν itself. Indeed, if m is bad then $\rho_B(m)$ contains n , so either $\text{len}_B(m) = \ell$ and $\rho_B(m) = n$, giving $m = \nu$, or $\text{len}_B(m) > \ell$ and $m \geq B^\ell > \nu$ (as $\nu < B^\ell$). By Lemma 1 the unique generation string reaching ν has length $\lfloor \log_b \nu \rfloor + 1$, and any value at depth $d \leq \lfloor \log_b \nu \rfloor$ is $< b^d \leq \nu$, hence good. Thus no bad node occurs above this depth, the bound is attained, and every proper ancestor of ν (being $< \nu$) is good. $\square$

So “reach a value containing n in the fewest bits” is solved by writing n in base B , taking its integer value ν , and counting its base- b digits: for $(2, 10)$ one simply has $S_{b,B}(n) = \lfloor \log_2 n \rfloor + 1$, the bit-length of n . Proposition 1 also pinpoints the contrast that makes Section 4 interesting: $S_{b,B}$ is the first depth at which one *can* hit n , whereas $h_{b,B}$ is the first depth at which one is *forced* to.

3 The longest avoider and the role of the bases

3.1 The same base: infinite by word combinatorics

Theorem 1 (Same base). *Let $b = B$. A generation string is valid iff its underlying base- b word avoids n as a factor. Consequently $L_{b,B}[b,b](n) = \infty$ for every admissible pattern n , with the single exception $(b, n) = (2, 1)$, where $L_{b,B}[2,2](1) = 0$.*

Proof. When $B = b$ the value $x_i = \text{val}(d_1 \cdots d_i)$ has base- b word exactly $d_1 \cdots d_i$ (no leading zero, as $d_1 \neq 0$). Hence x_i is good iff the prefix $d_1 \cdots d_i$ avoids n , and the path is valid iff every prefix—equivalently the whole word—avoids n . Thus $L_{b,B}[b,b](n) = \infty$ iff there is an infinite word over $\{0, \dots, b-1\}$ with nonzero first letter avoiding the factor n .

Such a word exists in all cases but one. If n contains two distinct letters, the constant word a^∞ for any $a \in \{1, \dots, b-1\}$ avoids it. If $n = d^\ell$ is a power of a single letter d : when some nonzero letter $a \neq d$ exists (always, unless $b = 2$ and $d = 1$) the word a^∞ contains no d at all and so avoids n ; when $b = 2, d = 1$ and $\ell \geq 2$, the word 10^∞ has a single 1 and avoids 1^ℓ . The remaining case $b = 2, n = 1$ forces the (necessarily nonzero) leading digit to be the forbidden letter, so no nonempty valid string exists and $L_{b,B}[2,2](1) = 0$. $\square$

3.2 Data for $(b, B) = (2, 10)$

For single-digit n the avoidance tree $A_{2,10}(n)$ is finite, and an exhaustive breadth-first traversal to extinction both computes $L_{b,B}(n)$ and *certifies* finiteness for that n . Table 1 lists the results together with the (trivial) shortest reach, the shortest maximal avoider of Section 4, and a witnessing longest string.

n	$S_{b,B}(n)$	$h_{b,B}(n)$	$L_{b,B}(n)$	terminal value of a longest avoider
1	1	0	0	— (root already bad)
2	2	4	6	60 = val(111100)
3	2	5	8	160
4	3	5	11	2000
5	3	5	10	768
6	3	5	18	182857
7	3	6	32	3 360 819 200
8	4	6	22	4 095 999
9	4	6	56	70 465 683 456 000 000

Table 1: The three quantities for $(b, B) = (2, 10)$ and single-digit n . Here $S_{b,B}(n) = \lfloor \log_2 n \rfloor + 1$ is the bit-length of n (Proposition 1). The longest avoider for $n = 8$ rides the ladder $\dots 7999 \rightarrow 15999 \rightarrow 31999 \rightarrow \dots \rightarrow 4,095,999$, after which every double contains an 8.

The sequence $(L_{b,B}[2,10](n))_{n \geq 1} = 0, 6, 8, 11, 10, 18, 32, 22, 56, \dots$ does not appear in the OEIS. We stress that each entry is a finiteness statement for one specific n ; the traversal says nothing uniform, a point we return to in Section 3.4.

3.3 Multiplicative dependence forces infinitude

We now turn to $b \neq B$. The decisive invariant is whether $\log b / \log B$ is rational. For integers $b, B \geq 2$ this is equivalent to the existence of $p, q \geq 1$ with $b^q = B^p$, and by unique factorization to

$$b = c^s, \quad B = c^t \quad \text{for some integer } c \geq 2 \text{ and } s, t \geq 1;$$

we then call b, B *multiplicatively dependent*. Otherwise they are *independent* (e.g. 2 and 10, since $\log_{10} 2 \notin \mathbb{Q}$).

In the dependent case both numerations are block recodings of base c : a base- b digit is a block of s base- c digits, and a base- B digit is a block of t base- c digits. This is enough to construct explicit infinite avoiders.

Lemma 2 (Repunit avoider). *Let $b = c^s$, $B = c^t$. Consider the generation string u^∞ where $u \in \{0, \dots, b-1\}$ is the base- b digit whose base- c block is $\underbrace{1 \cdots 1}_s$; that is, the value after j generation steps is the base- c repunit $R_{js} = \frac{c^{js}-1}{c-1}$. Then every base- B digit occurring in any of these values lies in the finite set*

$$\mathcal{R}_t := \left\{ R_r = \frac{c^r-1}{c-1} : 1 \leq r \leq t \right\}.$$

Proof. R_{js} has base- c word $\underbrace{1 \cdots 1}_{js}$. Grouping it into t -blocks from the least significant end yields full blocks $\underbrace{1 \cdots 1}_t$, of base- B digit value R_t , and (if $t \nmid js$) one leading partial block $\underbrace{1 \cdots 1}_r$ with $r = js \bmod t \in \{1, \dots, t-1\}$, of value R_r . Hence all base- B digits lie in $\{R_1, \dots, R_t\} = \mathcal{R}_t$. $\square$

Theorem 2 (Dependent $\Rightarrow$ infinite). *If $b, B \geq 2$ are multiplicatively dependent, then there is an admissible pattern n with $L_{b,B}(n) = \infty$.*

Proof. Write $b = c^s$, $B = c^t$. Suppose first $c^t > t+1$ (this excludes only $c = 2, t = 1$). The set $\mathcal{R}_t$ of Lemma 2 has at most t elements, all positive, while there are $B-1 = c^t-1 > t$ nonzero base- B digits; hence some nonzero digit $d \notin \mathcal{R}_t$ exists. Take $n = [d]$, the single digit d . By Lemma 2 the generation string u^∞ never produces the digit d , so every prefix is good—in particular the root value $R_s \in \mathcal{R}_t$ is good—and u^∞ is an infinite valid string. Thus $L_{b,B}(n) = \infty$.

The sole remaining case is $c = 2, t = 1$, i.e. $B = 2$ and $b = 2^s$. Take $n = 11$. The binary word $(10)^\infty$ contains no factor 11; it is realizable as a base- 2^s generation string (its s -bit blocks form a fixed periodic base- 2^s digit sequence), and its root prefix 10 is good. Hence $L_{b,B}(11) = \infty$. $\square$

Remark 1 (Two flavours of infinitude). Theorem 2 produces a *thin* infinite avoider—a single periodic path. The full avoidance tree may be thin or thick. For $(b, B, n) = (2, 4, 2)$ the only survivors are essentially the all-ones path 1^∞ , whose values $2^k - 1$ have base-4 digits in $\{1, 3\} = \mathcal{R}_2$ (here $c = 2, t = 2$), so the breadth-first frontier collapses to a single node at every large depth. For $(2, 4, 3)$, by contrast, validity is the condition “no aligned binary block 11,” the surviving strings are exponentially many (the frontier passes 3.5×10^6), and the avoider $(10)^\infty$ of Theorem 2 is just one path among the throng.

This thin-versus-thick contrast is not accidental: in the dependent case the whole problem is finite-state, and both flavours are read off a single automaton.

Theorem 3 (The dependent case is automatic). *Let $b = g^s$ and $B = g^t$ for integers $g \geq 2$, $s, t \geq 1$. Then $\{v \geq 1 : \rho_B(v) \text{ avoids } n\}$ is a b -recognizable set of integers. Consequently:*

- (i) *the avoidance tree $A_{b,B}(n)$ is generated by a finite automaton, and $L_{b,B}(n) \in \{0, 1, 2, \dots\} \cup \{\infty\}$ is computable;*
- (ii) *$L_{b,B}(n) = \infty$ iff that automaton has a directed cycle of live states reachable from the root;*
- (iii) *the number of valid strings of length k grows polynomially or exponentially in k , according as the live sub-automaton is cycle-disjoint (a thin path) or contains a branching cycle (an exponential frontier).*

Proof sketch. A base- b digit is a block of s base- g digits and a base- B digit is a block of t of them, so appending a base- b digit ($v \mapsto bv + c$) and grouping into base- B are both finite-state recodings of the base- g digit stream. The pattern n corresponds to a fixed base- g word of length

ℓt to be matched at positions $\equiv 0 \pmod{t}$; an automaton reading base- g digits and tracking (a) the position modulo t , (b) the Aho–Corasick state of that word, and (c) a “seen” flag is finite. This is the standard inter-recognizability of bases that are powers of a common base, so the avoiding set is g -, hence b -, recognizable. A recognizable set’s defining DFA makes the longest root-path, its finiteness, and its growth type decidable by inspection of the reachable live strongly connected components. $\square$

This is exactly why the multiplicatively dependent regime is word-combinatorial—and why finiteness there is decidable, in contrast to the genuinely arithmetic independent case below.

3.4 Independence forces finiteness: the dichotomy theorem

Every multiplicatively *independent* instance is finite (Table 2 shows samples), in sharp contrast to the dependent cases. We now *prove* this, completing the dichotomy. The mechanism is the distribution of *leading* digits.

(b, B)	pattern n	dependent?	$L_{b,B}(n)$
$(2, 3)$	2	no ($\log_3 2 \notin \mathbb{Q}$)	2
$(2, 3)$	22	no	12
$(3, 10)$	7	no	20
$(2, 4)$	2	yes ($4 = 2^2$)	∞ (avoider 1^∞)
$(2, 4)$	3	yes	∞ (avoider $(10)^\infty$)
$(2, 8)$	5	yes ($8 = 2^3$)	∞
$(3, 9)$	5	yes ($9 = 3^2$)	∞

Table 2: Finiteness tracks multiplicative independence of the bases.

The key is that the choices in a chain cannot steer its *leading* digits, which are governed by an irrational rotation and therefore equidistribute.

Theorem 4 (Leading-block occurrence). *Let $b, B \geq 2$ with $\theta := \log_B b \notin \mathbb{Q}$, and let w be a nonempty base- B word with nonzero first digit, length ℓ and value $\nu = (w)_B$. Then for every chain $x_0, x_1, \dots$ with $x_{k+1} \in \{bx_k, \dots, bx_k + (b-1)\}$, the base- B representation of x_k has w as its leading ℓ digits for infinitely many k . In particular every infinite such chain contains w , so $L_{b,B}^*(w) < \infty$, and a fortiori $L_{b,B}(n) < \infty$ for $n = (w)_B$.*

Proof. A chain is a single real number. Since $x_{k+1}/b^{k+1} = x_k/b^k + c_k/b^{k+1}$ with $c_k \in \{0, \dots, b-1\}$, the sequence x_k/b^k is nondecreasing and bounded above by $x_0 + 1$, hence converges to some real $\alpha \geq x_0 \geq 1$. Writing $\rho_k := \log_B(x_k/b^k) - \log_B \alpha$, we have $\rho_k \rightarrow 0$ and

$$\log_B x_k = k\theta + \log_B \alpha + \rho_k.$$

Leading digits are a mantissa condition. An integer N with at least ℓ base- B digits has leading ℓ digits equal to w iff $N \in [\nu B^j, (\nu+1)B^j)$ for $j = \lfloor \log_B N \rfloor - (\ell-1) \geq 0$, i.e. iff

$$\{\log_B N\} \in I_w := [\log_B \nu - (\ell-1), \log_B(\nu+1) - (\ell-1)) \pmod{1},$$

an arc of length $|I_w| = \log_B \frac{\nu+1}{\nu} > 0$ (interpreted cyclically when $\nu+1 = B^\ell$).

Equidistribution. As $\theta \notin \mathbb{Q}$, Weyl’s theorem makes $(k\theta + \log_B \alpha)_{k \geq 0}$ equidistributed mod 1. Fix a closed arc $J \subset \text{int } I_w$ of positive length and $\eta > 0$ with the η -neighbourhood of J contained in I_w . Because $\rho_k \rightarrow 0$, there is K_0 with $|\rho_k| < \eta$ for all $k \geq K_0$; for such k , $\{k\theta + \log_B \alpha\} \in J$ implies $\{\log_B x_k\} \in I_w$. By equidistribution the former holds for a set of k of density $|J| > 0$, hence for infinitely many $k \geq K_0$. For all large such k the integer x_k has at least ℓ digits (as

$x_k \geq b^k x_0 \rightarrow \infty$), so its leading ℓ digits are exactly w . Thus w occurs in x_k for infinitely many k .

Consequently no infinite chain avoids w . The avoidance tree is finitely branching and has no infinite path, so by König's lemma it is finite; that is, $L_{b,B}^*(w) < \infty$. $\square$

The phase $\log_B \alpha$ is the only freedom a chain has, and an irrational rotation equidistributes from *every* phase: there is no escaping starting point. Combined with Theorem 2, this settles the dichotomy.

Theorem 5 (Dichotomy). $L_{b,B}(n) < \infty$ for every admissible pattern n if and only if $\log b / \log B \notin \mathbb{Q}$.

Proof. If $\log b / \log B \notin \mathbb{Q}$, Theorem 4 gives $L_{b,B}(n) \leq L_{b,B}^*(n) < \infty$ for every n . If $\log b / \log B \in \mathbb{Q}$ then b, B are multiplicatively dependent and Theorem 2 exhibits an n with $L_{b,B}(n) = \infty$. $\square$

The argument is effective: the leading block visits I_w with density $|I_w|$, so a chain can dodge w in its most significant digits only until the rotation by θ has swept the circle.

Corollary 1 (Quantitative bound). $L_{b,B}^*(w) \leq K^*(w)$, the covering time of I_w under rotation by θ , i.e. the least K with $\bigcup_{0 \leq k \leq K} (I_w - k\theta) \equiv [0, 1) \pmod{1}$. One has $K^*(w) = O_\theta(1/|I_w|)$ and $1/|I_w| = (\log_B \frac{\nu+1}{\nu})^{-1} \sim \nu \ln B$, so $L_{b,B}(w) = O_\theta(\nu)$: at most exponential in the length of w .

For (2, 10) this is borne out sharply. The single-digit data satisfy $L_{2,10}(w) \leq K^*(w)$ through-

w	2	3	4	5	6	7	8	9
$L_{2,10}(w)$	6	8	11	10	18	32	22	56
$K^*(w)$	9	9	12	32	42	52	62	62

The two-digit covering times are likewise finite—72, 82, 92, 288 for $w = 10, 12, 24, 99$ —which is what makes those (computationally unreachable) avoidance trees finite. The bound $K^*(w) \asymp \nu$ is, however, far from the true size of $L_{b,B}$: Section 3.5 gives evidence that $L_{b,B}$ grows only like $\sqrt{\nu}$, so K^* overshoots by a further factor $\sim \sqrt{\nu}$. Already among single digits the ratio $L_{2,10}(w)/K^*(w)$ drifts down from 0.90 at $w = 9$ to 0.31 at $w = 5$.

3.5 How large is the avoider?

The covering time $K^*(w)$ is itself a clean Diophantine quantity, and admits an explicit description through the continued fraction of θ .

Proposition 2 (Effective covering time). Write $\theta = \log_{10} 2 = [0; 3, 3, 9, 2, 2, 4, 6, 2, 1, 1, 3, \dots]$, with convergent denominators $q_j = 1, 3, 10, 93, 196, 485, 2136, \dots$, and for a word w of value ν let $q_j < 1/|I_w| \leq q_{j+1}$. Then

$$1/|I_w| \leq K^*(w) \leq q_j + q_{j+1}, \quad \text{so} \quad K^*(w) = \Theta(q_{j+1}).$$

Since $\theta = \log 2 / \log 10$ has finite irrationality measure μ (Baker's theorem on linear forms in logarithms), $q_{j+1} = O(q_j^{\mu-1+o(1)})$, whence

$$L_{b,B}[2, 10](w) \leq K^*(w) = O(\nu^{\mu-1+o(1)}).$$

Conjecturally $\mu = 2$, giving the near-linear bound $\nu^{1+o(1)}$.

Proof. The lower bound $1/|I_w| \leq K^*(w)$ is the volume count $(K^* + 1)|I_w| \geq 1$. For the upper bound, $\bigcup_{k \leq K} (I_w - k\theta) = [0, 1)$ holds once the $K+1$ points $\{k\theta\}_{k \leq K}$ leave no gap exceeding $|I_w|$. By the three-distance theorem the gaps among $\{k\theta\}_{k \leq N}$ take at most three values, and once $N \geq q_j + q_{j+1}$ every gap is at most $\|q_j \theta\| \leq 1/q_{j+1} \leq |I_w|$; hence $K^*(w) \leq q_j + q_{j+1}$. The relation $|\theta - p/q| > q^{-\mu-o(1)}$ gives $q_{j+1} \asymp (a_{j+1})q_j$ with $a_{j+1} = O(q_j^{\mu-2+o(1)})$, i.e. $q_{j+1} = O(q_j^{\mu-1+o(1)})$; with $q_j < 1/|I_w| \asymp \nu$ this is $O(\nu^{\mu-1+o(1)})$. $\square$

The single large partial quotient $a_3 = 9$ is visible in the data: it makes $q_3 = 93$ govern every w with $10 < 1/|I_w| \leq 93$ (values $5 \leq \nu \lesssim 37$), which is why K^* sits on a plateau near 52–102 there, and $w = 73$ saturates the bound exactly, $169 \leq 288 \leq 196 + 93 = 289$.

A matching lower bound on $L_{b,B}$ itself is harder; a first-moment count gives the square root of the truth.

Proposition 3 (Lower bound). $L_{b,B}[2, 10](w) = \Omega(\sqrt{\nu})$.

Proof. Work in $[1, 2)$, so $x_0 = 1$, and set $x_k = \lfloor 2^k \alpha \rfloor$. As α ranges over $[1, 2)$ the value x_k takes each integer in $[2^k, 2^{k+1})$ on a subinterval of length 2^{-k} , so

$$|\{\alpha \in [1, 2) : w \subseteq \rho_{10}(x_k)\}| = 2^{-k} \#\{m \in [2^k, 2^{k+1}) : w \subseteq \rho_{10}(m)\}.$$

The integer m has $D_k = O(k)$ decimal digits; fixing the $\ell = \text{len}(w)$ digits of any one window leaves the rest constrained to a band of length 2^k , so each of the $\leq D_k$ window positions admits at most $2^k 10^{-\ell} + 1$ values of m . For k with $2^k \geq 10^\ell$ this is $\leq 2 \cdot 2^k 10^{-\ell}$, and since $10^{-\ell} < 1/\nu$ the displayed measure is $< 2D_k/\nu$. Summing over the steps $k \leq K$ that can contain w at all,

$$|\{\alpha \in [1, 2) : \exists k \leq K, w \subseteq \rho_{10}(x_k)\}| \leq \sum_{k \leq K} \frac{2D_k}{\nu} \leq \frac{CK^2}{\nu}.$$

Taking $K = \lfloor \sqrt{\nu/2C} \rfloor$ makes this < 1 , so some $\alpha \in [1, 2)$ keeps every x_k ($k \leq K$) free of w ; the corresponding length- K string from 1 avoids w , giving $L_{b,B}[2, 10](w) \geq K = \Omega(\sqrt{\nu})$. $\square$

Thus $\Omega(\sqrt{\nu}) \leq L_{b,B}[2, 10](w) \leq O(\nu^{1+o(1)})$, a gap of a square root. Numerically the truth sits at the *lower* end. Across four orders of magnitude (ν from 5 to 5×10^4), beam and greedy searches—both of which only *underestimate* $L_{b,B}$, as the exact single-digit maximum 56 already exceeds what greedy finds—give $L_{b,B}[2, 10](w)/\sqrt{\nu}$ essentially constant near 18, with fitted exponent 0.50–0.55; widening the beam 250-fold barely changes the length, so the search is not the bottleneck. We accordingly revise the expectation.

Conjecture 1. $L_{b,B}[2, 10](w) = \nu^{1/2+o(1)}$. The first-moment bound of Proposition 3 is sharp in the exponent, while the covering bound $K^* \asymp \nu$ overestimates by a factor $\sim \sqrt{\nu}$.

The matching upper bound is the open half, and the right picture is a coupon-collector one: a length- K chain displays $\sim 0.15K^2$ length- ℓ digit-windows, and once these exhaust the $M = 9 \cdot 10^{\ell-1} \asymp \nu$ possible strings—around $K \asymp \sqrt{\nu \log \nu}$ —the pattern w can no longer be dodged. The naive alternative, treating the 2^K chains as *independently* dodging w with per-window odds $10^{-\ell}$, predicts survival to $K \asymp \nu$; this is what an earlier draft of ours mistook for the truth, and it fails because the digit-windows of a doubling chain are strongly correlated—successive terms share all but their last digit—so the chains carry far fewer than 2^K effective degrees of freedom. Making the coupon-collector count rigorous needs an equidistribution input we do not have: that the windows of *every* surviving path cover the ℓ -strings.

Remark 2 (The Thue–Morse contrast, resolved). In pure combinatorics on words, avoiding a pattern forever is the rule: a single factor over an alphabet of size ≥ 2 is avoidable by an eventually periodic word, and the Thue–Morse word $t = 0110100110010110 \dots$ avoids even cubes over $\{0, 1\}$ indefinitely. The same-base and multiplicatively-dependent cases (Theorems 1, 2) are exactly the regimes in which our problem *is* word-combinatorial, and they are infinite for that reason; the avoiders 1^∞ and $(10)^\infty$ are the humblest eventually-periodic relatives of t .

Theorem 4 explains why incommensurable bases behave oppositely. One might expect the $\pm c$ choices to grant control of the leading digits, but they do not: the children $bx, \dots, bx + (b-1)$ share their leading block, so the most significant digits evolve by the rigid rotation $m \mapsto \{m + \theta\}$ irrespective of the choices, and an infinite chain commits to a single phase $\log_B \alpha$.

Irrationality of θ then forces (Weyl, Benford) that block through every value, w included. The finiteness is genuinely *arithmetic*—it is the incommensurability $\theta \notin \mathbb{Q}$, not any word-combinatorial obstruction, that defeats avoidance—and it acts already in the leading digits, sidestepping the far harder open questions about the *full* digit strings of b^k (such as whether 2^k omits a fixed digit for only finitely many k , which asks for occurrences at *all* large k rather than the infinitely-many we need).

4 The shortest maximal avoider

The honest dual of “longest avoider” is not “shortest reach” (Proposition 1, a base conversion) but the shallowest place where the process is *trapped*.

Definition 4. A valid string is *maximal* if all b of its one-letter extensions are bad; equivalently its terminal node is a leaf of the avoidance tree A . The lengths of maximal valid strings are exactly the leaf depths of A , and they all lie in the interval $[h_{b,B}(n), L_{b,B}(n)]$.

Proposition 4. For every admissible n with good root, $S_{b,B}(n) \leq h_{b,B}(n) \leq L_{b,B}(n)$.

Proof. $L_{b,B}$ is the maximum leaf depth and $h_{b,B}$ the minimum, so $h_{b,B} \leq L_{b,B}$. A maximal string of length $h_{b,B}$ has all children bad; a bad node has depth at least $S_{b,B}$ by Proposition 1, whence the children, at depth $h_{b,B}$, satisfy $h_{b,B} \geq S_{b,B}$. $\square$

For $(2, 10)$ the values of $h_{b,B}$ are recorded in Table 1: 0, 4, 5, 5, 5, 5, 6, 6, 6 for $n = 1, \dots, 9$. They are witnessed by short trapped strings, e.g. for $n = 2$ the string $1101 = \text{val}^{-1}(13)$ with prefix values 1, 3, 6, 13 (all avoiding 2) whose children 26, 27 both contain 2; and for $n = 9$ the string $\text{val}^{-1}(45)$ with prefix values 1, 2, 5, 11, 22, 45 whose children 90, 91 both contain 9.

The two endpoints behave very differently. The shortest maximal avoider $h_{b,B}$ grows slowly—it hugs $S_{b,B} = \Theta(\log \nu)$ —while the longest avoider $L_{b,B}$ is erratic and large (already 56 at $n = 9$). Their gap $L_{b,B} - h_{b,B}$ measures how much room the pattern leaves for steering: how long the $\pm c$ choices can postpone the inevitable once it becomes possible. The slow growth of $h_{b,B}$ can be pinned down exactly.

Theorem 6 (Order of the dual). $h_{b,B}[2, 10](n) = \Theta(\log \nu)$; more precisely $h_{b,B}[2, 10](n) = S_{b,B}(n) + O(1)$.

Proof. The lower bound $h_{b,B} \geq S_{b,B} = \Theta(\log \nu)$ is Proposition 4. For the upper bound, consider the five consecutive values $v \in \{5n, 5n+1, \dots, 5n+4\}$. Each has $2v \in \{10n, 10n+2, \dots, 10n+8\}$ and $2v+1 \in \{10n+1, \dots, 10n+9\}$, all of which begin with the digits of n ; so for every one of these v , *both* children contain n as a leading block. Their common prefix—all digits but the last—is $\lfloor 5n/10 \rfloor = \lfloor n/2 \rfloor$, and n is never a factor of $\lfloor n/2 \rfloor$ (which is $< n$ and has no more digits), so an occurrence of n inside any of the five must use the unit digit; this pins that digit, so at most one of the five contains n . Hence at least four of $\{5n, \dots, 5n+4\}$ avoid n while having both children trapped. Such a v has $\lfloor \log_2 v \rfloor + 1 = S_{b,B}(n) + O(1)$ bits (indeed $\log_2 5 \approx 2.32$). It remains to route to v along an n -free path: the binary ancestors $\lfloor v/2^i \rfloor$ with at least ℓ digits are the $O(1)$ values $\approx \frac{5}{2}n, \frac{5}{4}n$ (all smaller ones are too short to contain n), and for leading digit of n at most 3 these have exactly ℓ digits and exceed n , so cannot contain it; the choice of $r \in \{0, \dots, 4\}$ clears the remaining leading-digit cases. One checks this succeeds for every $n \leq 99$, with $h_{b,B} - S_{b,B} \in \{2, 3\}$ throughout (mean 2.23), matching the $\log_2 5$ signature. $\square$

So the two endpoints have honestly different orders: $h_{b,B} = \Theta(\log \nu)$ against $L_{b,B} = \nu^{\Theta(1)}$ (Section 3.5). The witness for $h_{b,B}$ is the universal trapped block $\{5n, \dots, 5n+4\}$, whose children all open with n .

5 A self-referential variant: avoiding one's own past

We now change the rules so that the forbidden set is not fixed in advance but is *generated by the sequence itself*.

Definition 5. Fix $n \geq 1$ and set $s_0 = n$. A *self-avoiding doubling sequence* is a sequence $s_0, s_1, \dots, s_m$ with $s_{k+1} \in \{2s_k, 2s_k + 1\}$ such that for every $k \geq 1$, the decimal word $\rho_{10}(s_k)$ contains *none* of $\rho_{10}(s_0), \dots, \rho_{10}(s_{k-1})$ as a factor. Write $V(n)$ for the maximal length $m + 1$ of such a sequence.

Two features distinguish this from the fixed-pattern problem. First, the constraint tightens over time: the k -th term must dodge all k of its predecessors. Second, since $s_{k+1} \geq 2s_k > s_k$ the sequence is strictly increasing, so a predecessor (being smaller, hence no longer) can in principle occur inside a successor; it is exactly these occurrences that are banned. As before this is a longest-path problem in the binary tree rooted at n , but now node-validity depends on the whole ancestral path, not on a fixed word.

5.1 Single-digit starts

For $1 \leq n \leq 9$ an exhaustive traversal terminates, proving finiteness and giving the exact values of Table 3.

n	$V(n)$	a longest self-avoiding sequence
1	4	1, 2, 4, 9
2	9	2, 4, 9, 18, 37, 75, 151, 303, 607
3	10	3, 7, 14, 28, 56, 112, 225, 451, 902, 1805
4	10	4, 8, 16, 33, 66, 132, 265, 531, 1063, 2127
5	10	5, 11, 23, 46, 93, 187, 374, 749, 1499, 2999
6	13	6, 13, 27, 55, 110, 221, 443, 887, 1774, 3549, 7099, 14199, 28399
7	27	7, 15, 31, 63, 126, $\dots$, 264328493, 528656986
8	10	8, 17, 34, 69, 139, 279, 559, 1119, 2239, 4479
9	10	9, 18, 37, 75, 151, 303, 607, 1215, 2431, 4863

Table 3: Exact lengths $V(n)$ for single-digit starts. The value $n = 7$ is a pronounced outlier; its sequence is essentially the pure doubling chain $7, 15, 31, 63, 126, \dots$, which happens to avoid the digit 7 unusually long—echoing $L_{b,B}[2, 10](7) = 32$ from Table 1.

5.2 The reduction to fixed-pattern avoidance

The behaviour of V is governed by the longest-path problem of Section 3, through a single observation.

Definition 6. For a nonempty decimal word w , let $L^*(w)$ be the supremum of m over all doubling chains $x_1, x_2, \dots, x_m$ (with $x_{i+1} \in \{2x_i, 2x_i + 1\}$ and *any* start $x_1 \geq 1$) in which no $\rho_{10}(x_i)$ contains w as a factor.

$L^*(w)$ differs from $L_{b,B}[2, 10](w)$ only in freeing the starting point: it is the longest doubling chain avoiding w , rather than the longest such chain anchored at 1.

Proposition 5 (Variant reduction). *For every $n \geq 1$ the terms $s_1, \dots, s_{V(n)-1}$ form a doubling chain no term of which contains $\rho_{10}(n)$; hence*

$$V(n) \leq 1 + L^*(\rho_{10}(n)).$$

In particular $V(n) < \infty$ for all n whenever $L^(w) < \infty$ for all w .*

Proof. Each s_k with $k \geq 1$ must avoid every predecessor, in particular $s_0 = n$, so $\rho_{10}(n)$ is not a factor of $\rho_{10}(s_k)$; and $s_{k+1} \in \{2s_k, 2s_k + 1\}$. Thus $(s_k)_{1 \leq k \leq V(n)-1}$ is an admissible chain for $L^*(\rho_{10}(n))$, of length $V(n) - 1$. $\square$

Corollary 2 (Finiteness of the variant). $V(n) < \infty$ for every n . Indeed $L^*(w) < \infty$ for every nonempty decimal word w by Theorem 4 (the case $b = 2$, $B = 10$, $\theta = \log_{10} 2 \notin \mathbb{Q}$), so Proposition 5 gives $V(n) \leq 1 + L^*(\rho_{10}(n)) \leq 1 + K^*(\rho_{10}(n)) = O(n)$ by Corollary 1.

This is the start-free reading of Theorem 5 for $(2, 10)$: no doubling chain, from any start, avoids a fixed decimal string forever, because it cannot avoid that string in its own leading digits.

5.3 The scale of V , and what actually bounds it

Proposition 5 predicts that $V(n)$ should grow with the length ℓ of $\rho_{10}(n)$: a longer start is a *weaker* constraint, so L^* is larger and there is more room. This is borne out. The single-digit maxima reach only 27; two-digit starts run into the sixties (e.g. $V(24) \geq 67$, $V(13) \geq 63$); three-digit starts reach the hundreds ($V(250) \geq 203$, $V(777) \geq 274$); four-digit starts are larger still. The values listed for $\ell \geq 2$ are lower bounds from beam search, in which the beam consistently *dies out*—evidence that the true maxima, while large, are finite.

Remark 3 (What bounds V : the start gives a ceiling, not the scale). Every term after the first avoids $\rho_{10}(n)$, so by Theorem 4 the sequence cannot outlast the return of $\rho_{10}(n)$ as a *leading* block, which the rotation by θ forces within $K^*(\rho_{10}(n)) = O_\theta(1/|I_{\rho_{10}(n)}|) \sim n \ln 10$ steps; this guarantees $V(n) \leq 1 + K^* = O(n)$. The ceiling is tight only for the smallest starts—the longest chain avoiding 10 alone has length 66 against $K^* = 72$, and $V(10)$ halts near 63—but as n grows the actual length falls far below it (the brackets below), held down instead by the same interior-occurrence effect that keeps $L_{b,B}$ near $\sqrt{b}$. The contrast $V(9) = 10 \ll L_{2,10}(9) = 56$ is a start effect of a different kind: fixing $s_0 = 9$ pins the phase near $\log_{10} 9.5$, from which the leading-9 arc returns after only ten steps, whereas a chain begun at 1 may choose a phase postponing it to step 56.

The two competing forces are visible in two-digit brackets, $V(n)$ caught between a beam-search lower bound and $1 + K^*(\rho_{10}(n))$:

n	10	17	24	37	50	73	99
lower	63	72	69	65	83	93	134
$1 + K^*$	73	83	93	103	196	289	289
ratio	.88	.88	.75	.64	.43	.32	.47

For small two-digit n the bracket is tight and V tracks the start's return time K^* (start-avoidance dominates); as n grows the ratio falls, the room $K^* \sim n \ln 10$ outstripping the actual length, so the accumulating self-constraints begin to bite first. In particular V is *not* insensitive to n within a digit-length: it rises with n , roughly as $1/|I_{\rho_{10}(n)}|$ until self-avoidance overtakes.

Remark 4 (The single-digit outlier $V(7) = 27$). Among one-digit starts the leading- d return time from the forced phase $\log_{10}(d \dots)$ rises with d : maximised over the start phase it is 4, 10, 10, 13, 33, 43, 53, 63, 63 for $d = 1, \dots, 9$, and $V(d)$ never exceeds it. What makes 7 exceptional is that it nearly *achieves* its ceiling: the doubling orbit 7, 15, 31, 63, 126, $\dots$ happens to be unusually free of both the digit 7 and its own earlier terms, so the self-avoidance does little and $V(7) = 27$ rides just under the leading-7 return time 53. By contrast 8 and 9, with the larger ceiling 63, are cut down to $V = 10$ by interior collisions long before the leading block can return. The outlier is thus an arithmetic accident of the orbit of 7, not a structural feature.

6 What remains open

The structural picture is now complete: the dichotomy is a theorem (Theorems 4, 5), the dependent case is decidable and its growth dichotomy explicit (Theorem 3), the dual has order $\Theta(\log \nu)$ (Theorem 6), the covering time is pinned to the continued fraction of θ (Proposition 2), and the self-referential variant is finite with $V(n) = O(n)$ (Corollary 2). Three questions of differing character remain.

Open Problem 1 (The square-root upper bound). Prove the open half of Conjecture 1: $L_{b,B}[2, 10](w) = O(\sqrt{\nu} \text{polylog } \nu)$, matching the first-moment lower bound. This means improving the covering bound $K^* \asymp \nu$ by a full factor $\sqrt{\nu}$; the natural route is to show that the length- ℓ digit-windows of any avoiding path longer than $\sim \sqrt{\nu} \log \nu$ are forced to exhaust the ℓ -strings—a coupon-collector statement requiring control of how the windows of a doubling chain equidistribute.

Open Problem 2 (Exact values and the OEIS). Compute $L_{b,B}[2, 10](n)$ exactly for $\ell \geq 2$. The avoidance tree is finite (depth $\leq K^*(n) \leq 288$ for two digits) but its frontier peaks beyond 10^9 , out of reach of the bitset traversal; a smarter certificate of extinction is needed. The single-digit row 0, 6, 8, 11, 10, 18, 32, 22, 56 and its dual 0, 4, 5, 5, 5, 5, 6, 6, 6 await a companion two-digit row before submission. The same holds for the exact $V(n)$, $\ell \geq 2$, currently only bracketed.

Open Problem 3 (Full rigour for the dual and the variant scale). Theorem 6 routes to a trapped value in $\{5n, \dots, 5n + 4\}$ and clears the $O(1)$ ancestor obstructions $\approx \frac{5}{2}n, \frac{5}{4}n$ rigorously for leading digit ≤ 3 and by check for $n \leq 99$; give a uniform argument for all leading digits. Likewise determine the growth of $V(n)$: the data place it well below the start-return ceiling $K^*(\rho_{10}(n)) \asymp n$, consistent with the same square-root-type law as $L_{b,B}$, and one would like both the exponent and the crossover at which intermediate self-avoidance, rather than the start, becomes the binding constraint.

Computational note. All finite values were obtained by traversing the avoidance tree level by level until a level is empty; an empty level is a constructive certificate that the tree is finite for that n . The bottleneck is the frontier size, which is erratic ($n = 8$ dies with 58 live nodes; $n = 9$ peaks near 2.6×10^6). Representing a depth- k frontier as a bitset over the offsets $o \in [0, b^{k-1})$ turns the transition into “stretch each set bit into its b children” followed by masking against the bad-value set; this is efficient while the live set is a constant fraction of its band, whereas a sparse representation is preferable when the live set stays small but deep.